

Simple Folder Encryption

Windows EFS (Encrypting File System) — Hands-On Assignment Report (Final, Verified)

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1. What Was the Goal?

To lock a folder using a security feature that is already built into Windows, called EFS (Encrypting File System) — so that no one else, not even someone using the same computer, can open the files inside without the right account.

2. What is EFS? (In Simple Words)

Think of EFS like a lock that is tied to your Windows login, not a password on the folder itself. When you turn it on, Windows scrambles the files using a strong secret key (AES-256). That key is then locked away using your own Windows user certificate. So:

- If you are logged in as yourself → files open normally, no extra password needed.
- If someone else logs into a different account on the same PC, or steals the hard drive and plugs it into another computer → the files are unreadable, useless data.

3. Step-by-Step — What We Did

Step 1 — Right-clicked the folder and opened Properties.

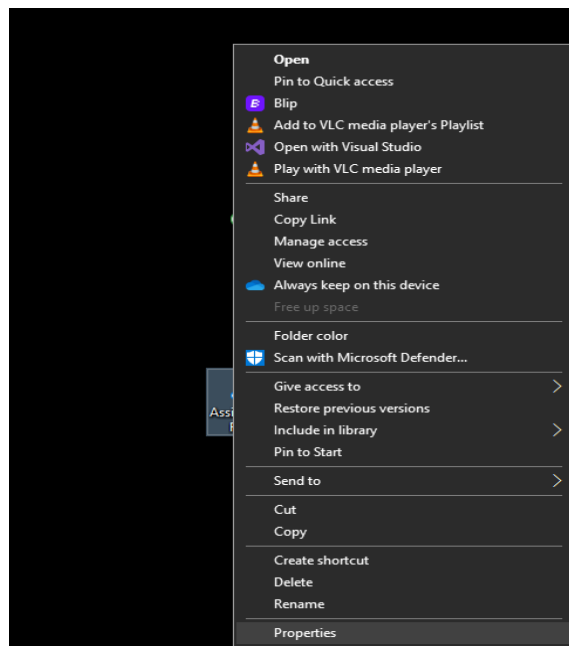


Fig 1. Right-click menu → Properties

Step 2 — Inside Properties (General tab), clicked the Advanced button.

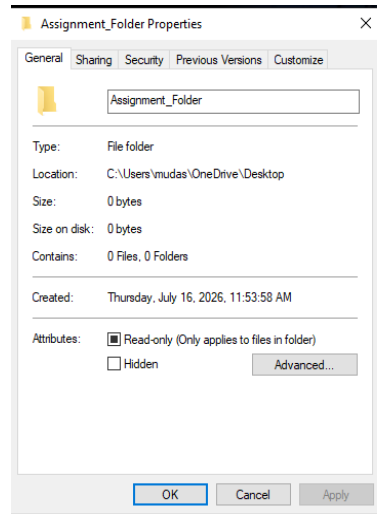


Fig 2. Properties window — General tab

Step 3 — The Advanced Attributes window opened. At first, 'Encrypt contents to secure data' was unchecked (encryption OFF).

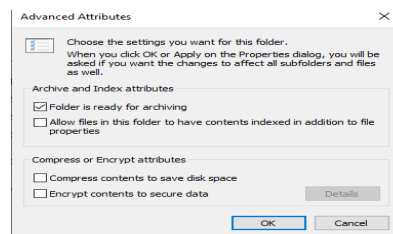


Fig 3. Advanced Attributes — before turning encryption on

Step 4 — Checked the encryption box and clicked OK.

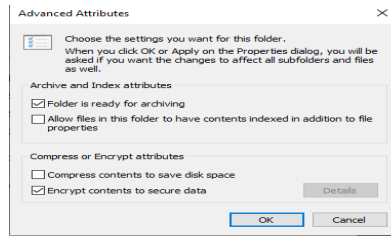


Fig 4. Advanced Attributes — encryption turned ON

Step 5 — Back in the Properties window, clicked Apply to actually start the encryption.

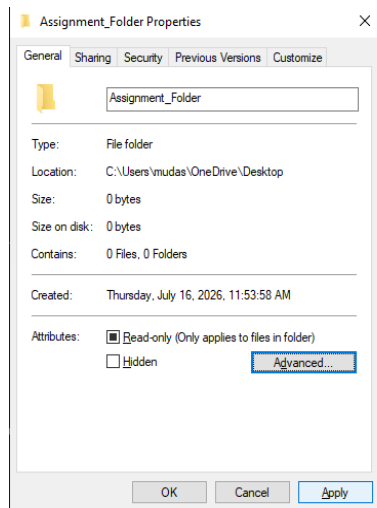


Fig 5. Properties window — clicking Apply

Step 6 — Windows asked: encrypt this folder only, or this folder + everything inside it? Chose 'this folder, subfolders and files' so nothing is left unprotected.

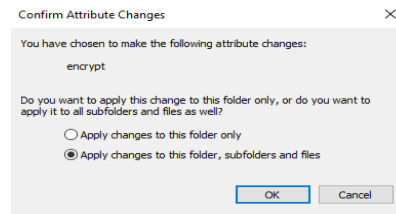


Fig 6. Confirm Attribute Changes — applying to all contents

Step 7 — Encryption finished, Explorer returned to normal view.

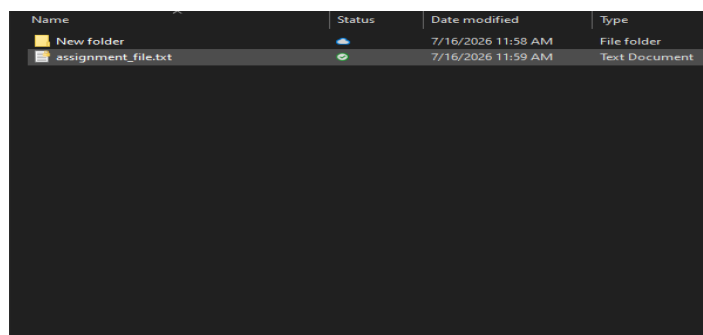


Fig 7. Explorer after encryption completed

Step 8 — Checked the folder icon. This is where something unexpected showed up — explained next.

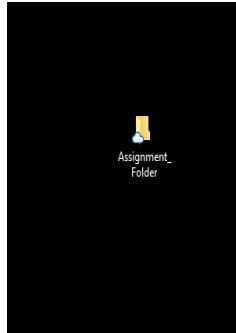


Fig 8. Folder icon — padlock symbol missing

4. Problem We Found (and Why It Matters)

Normally, once a folder is encrypted, Windows adds a small padlock icon on top of the folder picture, as visual proof. In Fig 8, that padlock is missing — even though the encryption steps were followed exactly right.

Reason: the folder was sitting inside OneDrive (a cloud backup/sync tool). OneDrive can interfere with how Windows shows — or even applies — EFS encryption, because it is constantly syncing files in the background.

This is a genuinely useful, real-world lesson: a security feature can look like it failed when actually it is a different tool (OneDrive) getting in the way. The fix is to never rely only on how something looks — always verify it properly.

5. The Fix — Moving the Folder + Verifying With a Command

Fix: moved the folder out of OneDrive to a plain local path: `D:\SOC_Projects\Assignment_Folder` — a location with no cloud sync involved.

Verification: instead of trusting an icon, used Windows' real encryption-check command:

```
cipher /c Assignment_Folder
```

```
Administrator: Command Prompt
Directory of D:\SOC_Projects
07/16/2026 12:09 PM <DIR>      .
07/16/2026 12:09 PM <DIR>      ..
07/16/2026 12:09 PM <DIR>      Assignment_Folder
                        0 File(s)      0 bytes
                        3 Dir(s)  31,476,555,776 bytes free

D:\SOC_Projects>cipher /c Assignment_Folder

Listing D:\SOC_Projects\
New files added to this directory will not be encrypted.

E Assignment_Folder
Compatibility Level:
  Windows XP/Server 2003

Users who can decrypt:
  KALI\mudas [mudas@mudas@KALI]
  Certificate thumbprint: F681 8B21 09EC EF58 C64A 2922 A0D9 4A8F 0588 CCDC

No recovery certificate found.

Key Information:
  Algorithm: AES
  Key Length: 256
  Key Entropy: 256

D:\SOC_Projects>
```

Fig 9. cipher /c output — folder confirmed ENCRYPTED

This command output is the real, trustworthy proof — much stronger than an icon. It shows:

- Status: E (Encrypted) — confirmed, not guessed
- Algorithm: AES, Key Length: 256 — strong, modern encryption standard
- Certificate thumbprint: F681 8B21 09EC EF58 C64A 2922 A0D9 4A8F 0588 CCDC
- Authorized user: KALI\mudas — proof that only this account can unlock it

6. A Common Mistake to Avoid

While checking, the same command was run again but got a strange, empty-looking result:

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.7417]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>cipher /c Assignment_Folder

Listing C:\Windows\system32\
New files added to this directory will not be encrypted.

C:\Windows\system32>
```

Fig 10. cipher /c run from the wrong folder — no real error, just wrong location

This was not an actual error. The command prompt was sitting inside C:\Windows\system32 at the time, and there is no folder named 'Assignment_Folder' there — so Windows had nothing to check. The lesson: always move into the correct folder first using `cd`, before running a command that checks something inside it.

```
cd /d D:\SOC_Projects cipher /c Assignment_Folder
```

7. What We Learned — Summary

- Encryption is already built into Windows — no extra software needed for basic file protection.
- EFS locks files to a specific Windows user account/certificate, not just a folder password.
- A visual icon (padlock) can be misleading — cloud-sync tools like OneDrive can hide or block it.
- The command line (cipher /c) gives the real, provable answer — this is how a professional verifies security settings instead of trusting appearances.
- Always double check which folder your terminal is in before running a command — an empty result can mean 'wrong location,' not 'it failed.'

8. Why This Matters in Real Life

This is exactly the kind of protection used on real company laptops. If a laptop with EFS-encrypted files is lost or stolen, whoever finds it cannot read those files — even by removing the hard drive and plugging it into another PC — because the encryption key is locked to the original Windows account. This directly supports Confidentiality, one of the three core pillars of cybersecurity (the CIA Triad: Confidentiality, Integrity, Availability).

9. Final Status

Item	Result
Folder encrypted?	Yes — confirmed via cipher /c (Status: E)
Encryption strength	AES, 256-bit key
Location	D:\SOC_Projects\Assignment_Folder (moved off OneDrive)
Authorized user	KALI\mudas
Assignment status	Complete — no further fixes needed

10. Next Phase (Optional Upgrade)

This assignment is now finished. As an optional next step, this can be turned into a small Python script that automatically runs the cipher /c check and prints a clean report — turning this GUI exercise into a real, code-backed project for GitHub.